

Micro-Motion Lie Detection



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Abstract

CRESCENT (Cognitive Response Evaluation via Subtle Change Enhancement and Neural Tracking) is an open-source, AI-powered system designed to non-invasively detect stress and potential deception by analyzing subtle physiological and behavioral cues from standard video and audio recordings. Combining motion amplification techniques like Eulerian Video Magnification (EVM), facial micro-expression analysis, voice pattern recognition, and machine learning classification, CRESCENT reveals imperceptible indicators such as pulse fluctuations, breathing irregularities, and facial muscular tension. The system operates in modular stages—including video preprocessing, motion enhancement, feature extraction, and classification—each using free, open-source technologies like OpenFace, MediaPipe, Yolor, Whisper.cpp, and scikit-learn. Designed to be scalable and adaptable, CRESCENT provides an alternative to invasive lie detection tools such as polygraphs and offers applications in fields like criminal investigation, psychological evaluation, security screening, and interview analysis. This paper outlines the conceptual framework and development roadmap for implementing CRESCENT in real-world scenarios, emphasizing its accessibility, ethical deployment, and potential to augment human judgment with data-driven insights.

1. Introduction

Can you detect lies based on micro-motion detection?

This query lies at the core of an Idea that combines computer vision, artificial intelligence, and behavioural science. In a world overwhelmed by misinformation, wrongful accusations, and deception—whether in courtrooms, police stations, airports, or online spaces—the ability to detect truth objectively is more critical than ever. Human perception, though intuitive, is fallible; traditional methods such as polygraphs are invasive and often unreliable. What if we could turn a standard video into a window revealing the body's hidden stress responses—without ever touching the subject?

This is the driving motivation behind CRESCENT (Cognitive Response Evaluation via Subtle Change Enhancement and Neural Tracking), a modular, AI-based lie detection framework that analyzes micro-movements and physiological signals—such as heart rate, respiration, and facial muscle twitches—amplified through Eulerian Video

Magnification (EVM). These signals, typically invisible to the human eye, provide measurable, repeatable clues about internal cognitive and emotional states.

CRESCENT has potential for mass use: psychological evaluation of non-verbal patients, enhanced security screening, telecommuting employment testing, forensic questioning analysis, and even courtroom testimony analysis. Such a tool is not simply technologically essential—it's ethically essential. In an era where truth can be concealed by bias, coercion, or manipulation, CRESCENT promises to offer forensic scientists, psychologists, and law enforcement a less biased and evidence-based instrument for understanding behavior.

By using only open-source tools and non-invasive techniques, this system stands as a low-barrier yet powerful innovation ; in a world of lies we need justices for the truth.

2. Background

2.1 Eulerian Video Magnification (EVM)

Developed by researchers at MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL), EVM is a system that amplifies the subtle variations in videos, such as tiny color changes and motion. The technique enables the visualization of physiological signals such as respiration and pulse from ordinary video footage. By amplifying the invisible signals, EVM provides the foundation for contactless physiological measurement, which is critical to apply in the case of lie detection where the application of physical sensors may not be feasible.

2.2 Facial Action Coding System (FACS) and OpenFace

The Facial Action Coding System (FACS) is a comprehensive system for coding human facial expressions in terms of muscle movements, or Action Units (AUs). OpenFace is open-source software that automates the analysis of facial behavior, including AU detection and head pose estimation, based on FACS. OpenFace has been instrumental in psychological research and human-computer interaction research in that it provides quantitative measurements of facial expressions that can be related to emotional and cognitive processes.

2.3 Whisper Speech Recognition

Whisper, an open-source automatic speech recognition system trained on a large multilingual and multitask supervised dataset created by OpenAI, demonstrates close-to-human robustness and accuracy in transcribing speech irrespective of accents, background noise, and technical jargon. Whisper's capacity to offer accurate transcriptions and temporally align them with audio waves renders it a solid building block for processing speech patterns and identifying anomalies that reflect deception.

2.4 Machine Learning with Scikit-learn

Scikit-learn is a free software machine learning library for the Python programming language that offers simple and efficient data mining and data analysis tools. It offers a range of supervised and unsupervised learning algorithms through a consistent interface, and building predictive models is simple. Scikit-learn allows CRESCENT to easily integrate several video and audio features for training classifiers to discern between deceptive and truthful behavior.

2.5 LibreFace: State-of-the-Art Facial Expression Recognition

LibreFace is a novel open-source deep facial expression analysis toolkit offering real-time and offline facial action unit detection, intensity estimation, and facial expression recognition. LibreFace relies on deep learning models and techniques like knowledge distillation and task-aware fine-tuning for superior performance and accuracy. LibreFace offers enhanced measurement of subtle facial expressions that are an important part of assessing emotional states and deception.

2.6 YOLOR

Object detection has undergone significant advancements, transitioning from traditional methods like R-CNN to more efficient models such as YOLO (You Only Look Once). YOLO reframes object detection as a single regression problem, enabling real-time detection by processing the entire image in one pass. By integrating YOLOR as a pre-processing step before using tools like OpenFace or LibreFace, the system ensures more accurate and stable tracking of facial features. This results in better extraction of Action Units (AUs), which enhances the reliability of downstream deception and stress detection algorithms.

3. Proposed System

CRESCENT (Cognitive Response Evaluation through Subtle Change Enhancement and Neural Tracking), is based on the hypothesis that involuntary micro-movements—such as subtle facial muscle contractions, blood flow fluctuations, irregular breathing patterns, and vocal irregularities—correlate strongly with internal cognitive load and psychological stress. These unconscious responses are particularly likely to surface when individuals are suppressing emotions, under scrutiny, or attempting to deceive. Our core idea is to detect and interpret these signals through non-invasive video and audio analysis, supported by machine learning—not through subjective observation, but through objective, quantifiable data.

CRESCENT will operate on a modular, multi-stage platform. It begins with a preprocessing module, in which ordinary video recordings (e.g., interviews, interrogations) are split into individual frames and

accompanied by isolated audio tracks. The next stage applies Eulerian Video Magnification (EVM) to amplify micro-scale color and motion changes. By focusing on frequency bands tied to respiration and heart rate, EVM reveals physiological details such as pulse-related color shifts in the face and minute muscular twitches that typically escape visual detection.

Concurrently, facial analysis tools such as OpenFace or LibreFace will be employed to detect Action Units (AUs)—standardized representations of facial muscle movements tied to emotional expressions. Whisper.cpp, an open-source speech recognition model, will transcribe and timestamp spoken content, capturing stress-related speech patterns like increased pauses, pitch shifts, or uneven pacing.

These visual and audio features will be compiled into a structured dataset. This dataset will be processed by machine learning models, such as Random Forests or LSTM neural networks, trained on labeled data identifying segments as

“truthful” or “deceptive.” The models will use these physiological and behavioral markers to estimate the probability of stress or deception at given moments in the footage.

Though the system is still in development, the planned output will be an interactive dashboard interface—likely built using tools like Streamlit. The dashboard will display synchronized playback of the enhanced video, heart rate graphs, AU activity, and flagged speech abnormalities. Risk scores and time-stamped highlights will allow investigators or clinicians to quickly focus on potentially significant moments.

CRESCENT is not yet a functioning program but represents a comprehensive design for an open-source, scalable, and non-invasive deception analysis platform. It has the potential to be deployed in fields ranging from law enforcement and clinical psychology to security and virtual interviewing, offering a new approach to behavioral truth assessment powered by technology. This program will be designed to improve through many trials by using machine learning to learning off of patterns of similar cases.

4. Experimental Procedure

The experimental design of CRESCENT (Cognitive Response Evaluation by Subtle Change Enhancement and Neural Tracking) will be utilized non-invasively to identify physiological and behavioral reactions to deception. The procedure consists of several steps:

4.1 Data Acquisition and Preprocessing

Video and Audio Extraction: FFmpeg is employed to decompose normal video recordings (e.g., interviews) into separate frames. The audio tracks corresponding to those are independently extracted for processing.

4.2 Motion Amplification

Eulerian Video Magnification (EVM): Employed to make visible subtle color and motion changes in some frequency band that are correlated with physiological signals like heart rate and respiration. EVM amplifies imperceptible changes, e.g., color changes in the skin due to blood flow.



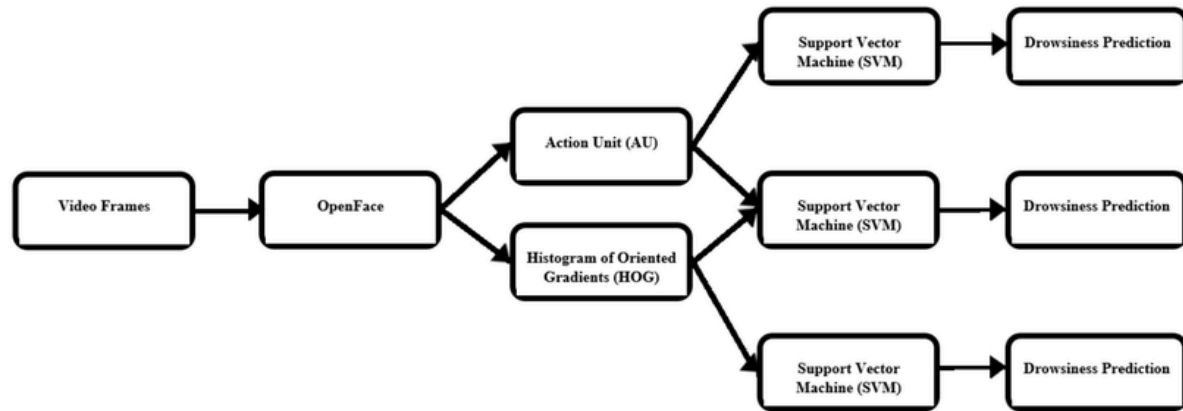
Visual Aid: demonstrations of EVM's application in contactless heart rate detection is available in the video "Contactless Heart Rate Detection using Eulerian Magnification":



4.3 Facial Feature Extraction

OpenFace Toolkit: Employed to detect facial landmarks and identify Facial Action Units (AUs) that are particular facial muscle movements indicating emotional expressions. OpenFace provides both static AU detection models and dynamic AU detection models with person-specific calibration.

visual Aid: An example of Action Units detected by OpenFace is presented in the following table:



5. Discussion

While CRESCENT is designed to be cloud-based and subscription-accessible, eliminating the obstacle of high local computing requirements, In *Truth* there remain several limitations—particularly in the accuracy and reliability of the system's core detection processes.

5.1: Environmental Sensitivity

interest is the environmental sensitivity of Eulerian Video Magnification (EVM). EVM augments small motion and color variations, such as blood flow or breathing, but is extremely sensitive to variables like video resolution, frame rate, and illumination levels. In low-light situations or compressed shots, signal-to-noise ratio decreases, potentially leading to false positives or loss of critical physiological signals. (However our target market is a niche market that can solidify these needs ,)

5.2 : Detection Through Diverse Pigmentation

A second temporary limitation pertains to variation in skin color. Since EVM and facial tracking computations rely on apparent color change in the red and green channels, individuals with darker skin tones or under different lighting will yield less accurate data. This introduces potential bias in the measurement of physiological change and could decrease model performance in a population with more varied demographics unless datasets and training are normalized across such variation. (That is why CRESCENT has many different ways of detecting lies and the sole premise is to detect lies however like all things accuracy is not 100 % because this program will be designed to improve through many trials by using machine learning to learning off of patterns of similar cases)

5.3: Interaction Of Multiple Systems

Facial landmark location and Action Unit (AU) analysis by software like OpenFace and LibreFace will also be affected by occlusion (e.g., glasses, facial hair), head movement, or non-frontal poses. While YOLOR is intended to increase robustness of facial detection, the interaction between multiple systems introduces complexity and possible mismatch between detected areas and physiological measurements of its own.

5.4 : Detection Through Diverse Dialects

Also, speech analysis based on Whisper.cpp could vary in performance depending on different accents, speech disabilities, or overlaid speech in noisy environments. In spite of cloud computing and mass processing, these deficiencies in the quality of inputs and inconsistency in human behavior have to be addressed to avoid incorrect deception tagging. (as i stated before this program will be designed to improve through many trials by using machine learning to learning off of patterns of similar cases)

Ultimately, with the good technological basis and cloud hosting removing limits on access, the equity and legitimacy of CRESCENT depend upon how well these edge cases are managed. Real-world testing under a range of circumstances, ethical analysis, and algorithmic auditing will be crucial to ensuring the system enables responsible and equitable truth determination. In my opinion, none of which I have stated are a limitation, for limitations do not exist. They are only temporary setbacks.

6. Conclusion & References

CRESCENT (Cognitive Response Evaluation through Subtle Change Enhancement and Neural Tracking) is a bold and audacious approach to deception detection and behavioral analysis. Based on the fundamental hypothesis that spontaneous involuntary physiological signals—micro-expressions, blood flow oscillations, respiration dynamics, and voice dysrhythmias—can be employed to measure emotional and cognitive distress, this system makes use of the strengths of computer vision, audio signal processing, and machine learning to transform normal video footage into rich sources of psychological data.

While there are limitations in lighting sensitivity, skin tone variation, and speech variation, the system is designed to get better. These are not taken as barriers but as shifting challenges to be overcome using real-world testing, feedback, and incremental machine learning. CRESCENT is not just a technological initiative—instead, it is a proposition of justice, transparency, and veracity in an ever-more covert world filled with misinformation and deception. By rendering imperceptible physiological responses conspicuous, and by contextualizing them with state-of-the-art AI, it allows professionals—psychologists and police officers, national security operatives, interviewers—to inform more prudent, equitable, and moral decisions. With proper investment, development, and validation, CRESCENT can set a new benchmark for behavioral analysis—one that is rooted in science, guided by ethics, and committed to revealing truth not through force or assumption, but through observation, amplification, and understanding to find we need justices for the truth.

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